

Risk Assessment



Project	Metal sheet packaging device	Product	Vacuum based packaging system
Date	2 May 2026		
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Work description	The system uses conveyor automation, manual bag placement, vacuum sealing, and heat sealing to package steel plates efficiently for safe transportation and storage.		

Risk Matrix

Likelihood \ Severity	Rarely (1)	Unlikely (2)	Possible (3)	Likely (4)	Highly Likely (5)	People	Assets
Very High (5)	5	10	15	20	25	Death	Severe Damage
High (4)	4	8	12	16	20	Major Injury	Major Damage
Medium (3)	3	6	9	12	15	Over 3 Day Injury	Moderate Damage
Low (2)	2	4	6	8	10	Under 3 Day Injury	Slight Damage
Very Low (1)	1	2	3	4	5	Minor Injury (No lost Time)	No Damage

Score	Priority	Action
1~4	Low	Monitor and maintain existing controls
5~9	Medium	Implement additional control measures where practical
10~15	High	Immediate corrective action required to reduce risk
16~25	Critical	Stop operation immediately until risk is eliminated or significantly reduced

#	Hazard Identified	Risk Description	Initial		Initial Score (L x S)	Mitigations	Final		Final Score (L x S)	Responsible Team
			L	S			L	S		
1	Forklift collision or load drop	Severe injury, Crushing, Equipment damage	2	5	10	Designated forklift operating zones, proper load securing procedures, and certified operator training should be implemented to minimize collision and load drop risks.	1	5	5	Logistics Team
2	Conveyor pinch/crush points	Hand entrapment, Crushing injury	3	4	12	Physical safety guards, emergency stop systems, and restricted operator access should be installed around conveyor moving parts to prevent entrapment injuries.	1	4	4	Mechanical Design Team

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			L	S			L	S		
3	Sharp steel plate edges	Cuts, lacerations	4	3	12	Protective edge covers, mandatory PPE gloves, and safe material handling procedures should be used to reduce cutting and laceration hazards.	2	2	4	Safety Team
4	Heat sealing burns	Thermal burns, Skin injury	3	3	9	Thermal insulation barriers, warning signage, and operator protective equipment should be provided to minimize burn injuries from hot sealing components.	1	3	3	Process Design Team
5	Product jam or blockage	Mechanical overload, Unsafe intervention	3	3	9	Automatic shutdown sensors, jam detection systems, and lockout procedures should be applied before manual intervention to reduce overload and injury risks.	1	3	3	Mechanical Design Team
6	Vacuum sealing failure	Packaging failure, Transport instability	3	2	6	Routine equipment inspections, seal quality verification, and preventive maintenance schedules should be maintained to ensure packaging reliability.	1	2	2	Quality Control Team
7	Ergonomic strain	Repetitive strain, Operator fatigue	4	2	8	Adjustable workstation design, task rotation, and ergonomic training should be implemented to reduce repetitive strain and operator fatigue.	2	2	4	Operations Team
8	Maintenance hazards	Burn injury, Entanglement, Accidental activation	2	4	8	Lockout/tagout procedures, machine power isolation, and maintenance-specific PPE should be enforced during servicing activities.	1	4	4	Maintenance Team

* L = Likelihood, S = Severity

* This risk assessment was conducted in accordance with SFS-EN ISO 12100:2010 principles